

Abnormally very high RAM usage since 1.13 (client side)

MC-188163 Performance

Status	Resolved / Fixed
Affected	1.15.2, 1.16 Pre-release 2, 1.16 Pre-release 3, 1.16 Pre-release 4, 1.16 Pre-release 6, 1.16 Pre-release 7 (+51 more)
Fixed in	1.19.4 Pre-release 1
Reported	2020-06-08

STEPS TO REPRODUCE

1. Open the Minecraft launcher and go to the "Installations" section.
2. Create or edit a profile, select version 1.12.2, and open "More Options".
3. In the "JVM Arguments" field, set the maximum memory argument to `-Xmx8G``.
4. Save the profile.
5. Open the operating system's task manager.
6. Launch Minecraft using the configured profile and wait until the game is fully loaded.
7. Observe the game's RAM usage in the task manager.
8. Repeat steps 2-7 for versions 1.13.2, 1.14.4, 1.15.2, and 1.16 Pre-Release 2, using the same `-Xmx8G`` argument.
9. Observe that RAM usage increases from approximately 450-500 MB in 1.12.2 to approximately 2.1 GB in 1.13.2 and approximately 5.0-5.2 GB in 1.14.4, 1.15.2, and 1.16 Pre-Release 2.

OBSERVED BEHAVIOUR

The Minecraft client uses substantially more RAM in versions from 1.13 onward than in 1.12.2 when given a sufficiently high heap limit. With the JVM argument set to `-Xmx8G`, observed usage was approximately 450 MB in 1.12.2, 2.1 GB in 1.13.2, 5.0 GB in 1.14.4, 5.2 GB in 1.15.2, and 5.2 GB in 1.16 Pre-Release 2. These measurements were taken with the game running on two different machines and with no other significant applications open apart from Discord and the recording software. The increased client memory usage begins with 1.13, which introduced DataFixerUpper, and remains present in the tested later versions. The client also showed progressively higher CPU usage between versions, including after loading had completed, particularly in 1.16 Pre-Release 2.

EXPECTED BEHAVIOUR

When Minecraft is launched with a generous JVM memory limit, the client should use a reasonable amount of RAM rather than rapidly consuming several gigabytes. Its memory usage should remain broadly comparable to 1.12.2, approximately 500 MB under the reported test conditions, instead of rising to about 2.1 GB in 1.13.2 and over 5 GB in 1.14.4-1.16 Pre-Release 2.

ENVIRONMENT

Minecraft Java Edition Versions: 1.15.2, 1.16 Pre-release 2, 1.16 Pre-release 3, 1.16 Pre-release 4, 1.16 Pre-release 6, 1.16 Pre-release 7, 1.16 Pre-release 8, 1.16 Release Candidate 1, 1.16, 1.16.1, 20w27a, 20w28a, 20w29a, 20w30a, 1.16.2 Pre-release 1, 1.16.2 Pre-release 2, 1.16.2 Release Candidate 1, 1.16.2 Release Candidate 2, 1.16.2, 1.16.3 Release Candidate 1, 1.16.3, 1.16.4 Pre-release 1, 1.16.4 Pre-release 2, 1.16.4 Release Candidate 1, 1.16.4, 20w46a, 20w48a, 20w49a, 21w03a, 1.16.5, 21w05a, 21w05b, 21w06a, 21w07a, 21w11a, 1.17, 1.17.1 Pre-release 1, 1.17.1, 21w37a, 21w41a, 21w44a, 1.18 Pre-release 1, 1.18 Pre-release 4, 1.18 Release Candidate 1, 1.18, 1.18.1, 22w11a, 22w13a, 22w16b, 1.19, 1.19.1 Release Candidate 1, 1.19.1 Pre-release 2, 1.19.1 Pre-release 4, 1.19.2, 22w45a, 1.19.3, 23w07a (includes snapshots)

ORIGINAL DESCRIPTION

The bug

Since 1.13, the RAM used by the game has increased drastically to reach more than 5 GB in 1.15.2 and higher when Minecraft is not limited by JVM default arguments (= put `-Xmx8G` for example)!

For comparison, 1.12.2 uses only 500 MB of RAM!

This problem is apparently due to the DataFixerUpper introduced in 1.13.

How to test

- Go to the "Installations" section of your Minecraft launcher.
- Click on the three dots that appear when you hover over the profile you want to use for this test, and then click "Edit". If you don't already have a profile, create one!
- Select the version you want to test.
- Click on "More Options"
- Change the "-Xmx" argument in the "JVM Arguments" section to allocate 6 GB of RAM or more. Example for 8 GB (recommended) : -Xmx8G
- Save your profile.
- Open the task manager of your operating system.
- Launch the game with the correct profile and observe the evolution of the RAM used by the game in the task manager.
- Repeat for each version you want to test.

Videos tests

Here are all the tests I did from 1.12.2 to 1.16 Pre-Release 2, on all stable versions of Minecraft.

I have allocated 8 GB of RAM each time so as not to restrict the game to too low a ceiling, which would not allow us to see the problem.

No other programs run on the PC (except Discord and my recording program).

I did the test on my two different machines and the results are the same.

- Minecraft 1.12.2 (~450 MB) : Client RAM Test 1.12.2.mp4
- Minecraft 1.13.2 (~2.1 GB) : Client RAM Test 1.13.2.mp4
- Minecraft 1.14.4 (~5.0 GB) : Client RAM Test 1.14.4.mp4
- Minecraft 1.15.2 (~5.2 GB) : Client RAM Test 1.15.2.mp4
- Minecraft 1.16 Pre-Release 2 (~5.2 GB) : Client RAM Test 1.16 Pre-2.mp4

We can also observe that with each new version, the CPU is used harder and harder, even after the game is fully loaded (especially in 1.16).

Note: This ticket is attached to

MC-190258

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which processes the server side.

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-188163>